

Engineering Mathematics Study Notes: D'Alembert's Ratio Test

1. Concept Identification

The notes introduce **D'Alembert's Ratio Test** (commonly known simply as the **Ratio Test**). This is a highly mechanical calculus test used to check if an infinite series converges or diverges. It is especially useful when the terms of a series contain exponents (like 2^n , 7^n) or factorials (like $n!$). Instead of comparing the series to another helper series, this test analyzes the growth behavior of the series by comparing each term directly to its immediate successor.

2. Mathematical Representations

The Core Limit Formula

Let $\sum u_n$ be an infinite series with positive terms. We analyze the behavior of the ratio of the next term (u_{n+1}) to the current term (u_n) as n grows infinitely large:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l$$

The Decision Threshold Rules

The value of the calculated limit l dictates the absolute final outcome based on three strict criteria:

If $l < 1 \implies$ The series is Convergent

If $l > 1 \implies$ The series is Divergent

If $l = 1 \implies$ The test is Inconclusive (Fail)

Important Note: If $l = 1$, the test provides no information, and a different strategy (like the Limit Comparison Test) must be selected.

3. The Algorithm (Procedural Recipe)

To solve an infinite series using the Ratio Test, execute this precise step-by-step procedure:

Step 1: Identify the Current Term (u_n)

Extract the main algebraic formula located inside the summation sign.

Step 2: Formulate the Next Term (u_{n+1})

Create a new expression by replacing every single occurrence of the letter n in the original formula with the quantity $(n + 1)$. Always wrap $(n + 1)$ in parentheses.

Step 3: Setup the Division Ratio Fraction

Write out the division limit structure. To make calculation easier, immediately convert the division into a multiplication problem by multiplying u_{n+1} by the reciprocal (flipped version) of u_n :

$$\frac{u_{n+1}}{u_n} = u_{n+1} \times \frac{1}{u_n}$$

Step 4: Break Down Exponents and Factorials

Use basic rules of algebra to split up combined exponent terms and expand larger factorials into smaller pieces so they can be canceled out easily.

Step 5: Force Factor Out Dominant n Terms

Pull out the highest powers of n from all remaining grouped brackets, cancel them, and compute the limit by substituting $n = \infty$.

Step 6: Evaluate Against Thresholds

Compare the calculated number l against the value 1 to conclude whether the series converges or diverges.

4. Operational Mechanics

The Core Algebraic Rules Needed

Before looking at the problems, you must know how the lecture breaks down complex terms mechanically:

Rule A: Splitting Exponents

An addition in an exponent is simply multiplication in disguise:

$$2^{n+1} = 2^n \times 2^1 = 2 \cdot 2^n$$

$$7^{n+1} = 7^n \times 7^1 = 7 \cdot 7^n$$

Rule B: Stripping Factorials

A factorial represents multiplying all whole numbers down to 1. You can strip off the first term to reveal the original factorial expression:

$$(n+1)! = (n+1) \times n \times (n-1) \times \cdots \times 1 = (n+1) \cdot n!$$

Solution to Problem 1: Growing Exponential Numerator

Test for convergence of the series: $\sum_{n=1}^{\infty} \frac{2^n}{n^3 + 1}$

Step 1: State the current term

$$u_n = \frac{2^n}{n^3 + 1}$$

Step 2: State the next term

Substitute $(n+1)$ into every n position:

$$u_{n+1} = \frac{2^{n+1}}{(n+1)^3 + 1}$$

Step 3: Setup the flipped multiplication limit

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{2^{n+1}}{(n+1)^3 + 1} \times \frac{n^3 + 1}{2^n} \right]$$

Step 4: Break down components and regroup terms

Split 2^{n+1} into $2^n \cdot 2$ and rearrange the matching components:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{2 \cdot 2^n}{2^n} \times \frac{n^3 + 1}{(n+1)^3 + 1} \right]$$

Cancel the 2^n terms:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[2 \times \frac{n^3 + 1}{(n+1)^3 + 1} \right]$$

Step 5: Factor out n^3 to compute the limit

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} &= \lim_{n \rightarrow \infty} \left[2 \times \frac{n^3 \left(1 + \frac{1}{n^3}\right)}{n^3 \left(1 + \frac{1}{n}\right)^3 + 1} \right] \\ \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} &= \lim_{n \rightarrow \infty} \left[2 \times \frac{n^3 \left(1 + \frac{1}{n^3}\right)}{n^3 \left[\left(1 + \frac{1}{n}\right)^3 + \frac{1}{n^3}\right]} \right] \end{aligned}$$

Cancel the n^3 terms outside the brackets:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[2 \times \frac{1 + \frac{1}{n^3}}{\left(1 + \frac{1}{n}\right)^3 + \frac{1}{n^3}} \right]$$

Substitute $n = \infty$, which turns all fractional components to 0:

$$l = 2 \times \frac{1 + 0}{(1 + 0)^3 + 0} = 2 \times 1 = 2$$

Step 6: Evaluate the threshold

$$l = 2 \implies 2 > 1$$

Conclusion: The series is divergent.

Solution to Problem 2: Growing Exponential Denominator

Test for convergence of the series: $\sum_{n=1}^{\infty} \frac{n^2}{7^n}$

Step 1: State the current term

$$u_n = \frac{n^2}{7^n}$$

Step 2: State the next term

$$u_{n+1} = \frac{(n+1)^2}{7^{n+1}}$$

Step 3: Setup the flipped multiplication limit

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{(n+1)^2}{7^{n+1}} \times \frac{7^n}{n^2} \right]$$

Step 4: Break down components and regroup terms

Split 7^{n+1} into $7^n \cdot 7$ and rearrange matching components:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{7^n}{7 \cdot 7^n} \times \frac{(n+1)^2}{n^2} \right]$$

Cancel the 7^n terms:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{1}{7} \times \left(\frac{n+1}{n} \right)^2 \right]$$

Simplify the internal fraction:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{1}{7} \times \left(1 + \frac{1}{n} \right)^2 \right]$$

Step 5: Compute the limit value

Substitute $n = \infty$, which turns the fractional component to 0:

$$l = \frac{1}{7} \times (1 + 0)^2 = \frac{1}{7} \times 1 = \frac{1}{7}$$

Step 6: Evaluate the threshold

$$l = \frac{1}{7} \implies \frac{1}{7} < 1$$

Conclusion: The series is convergent.

Solution to Problem 3: The Factorial Series

Using the Ratio Test, test the convergence of the series: $\sum \frac{n! \cdot 2^n}{n^n}$

Step 1: State the current term

$$u_n = \frac{n! \cdot 2^n}{n^n}$$

Step 2: State the next term

Carefully change all n variables to $(n + 1)$:

$$u_{n+1} = \frac{(n+1)! \cdot 2^{n+1}}{(n+1)^{n+1}}$$

Step 3: Setup the flipped multiplication limit

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{(n+1)! \cdot 2^{n+1}}{(n+1)^{n+1}} \times \frac{n^n}{n! \cdot 2^n} \right]$$

Step 4: Break down components to prepare for cancellation

- Expand the factorial: $(n+1)! = (n+1) \cdot n!$
- Expand the exponential base: $2^{n+1} = 2 \cdot 2^n$
- Expand the power base: $(n+1)^{n+1} = (n+1)^n \cdot (n+1)^1 = (n+1) \cdot (n+1)^n$

Substitute these expanded pieces back into our limit calculation line:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{(n+1) \cdot n! \cdot 2 \cdot 2^n}{(n+1) \cdot (n+1)^n} \times \frac{n^n}{n! \cdot 2^n} \right]$$

Cross out the matching pairs ($n!$, 2^n , and the standalone bracket term $(n+1)$):

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[2 \times \frac{n^n}{(n+1)^n} \right]$$

Consolidate the terms under a shared exponent power of n :

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[2 \times \left(\frac{n}{n+1} \right)^n \right]$$

Invert the internal fraction to move the power down to the denominator:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{2}{\left(\frac{n+1}{n}\right)^n} \right]$$
$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \left[\frac{2}{\left(1 + \frac{1}{n}\right)^n} \right]$$

Step 5: Compute the limit value using the special identity

Recall the fundamental calculus identity rule: $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$ (where Euler's constant $e \approx 2.718$).

$$l = \frac{2}{e}$$

Step 6: Evaluate the threshold

Substitute the value of $e \approx 2.718$ to check the size of the fraction:

$$l = \frac{2}{2.718} \approx 0.736 \implies 0.736 < 1$$

Conclusion: The series is convergent.